Experiment 32: Esters

Notes

The chemistry of pleasant aromas and flavours is very often associated with the chemistry of esters. Natural aromas from fruits and herbs are due to the production of esters by plants.

Many fruit essences used in the food industry contain the same esters found in fruit. Examples of these include esters responsible for the characteristic odours and flavours associated with fruits such as bananas, oranges and pineapples.

In addition to the fruit esters, many other esters are commonly encountered in our everyday lives. Aspirin (acetylsalicylic acid), oil of wintergreen (methyl salicylate) and ethyl ethanoate (a common solvent used, for example, in nail polish remover) are all examples of esters.

Esters are formed by the reaction of carboxylic acids with alcohols, usually in the presence of a catalyst such as sulfuric acid. This reaction is called esterification. For example ethyl ethanoate is formed by the reaction between acetic acid and ethanol.

In this experiment you will prepare several esters and to try and identify some you may have encountered previously.

Equipment

Bunsen burner, tripod and gauze mat
beakers (100 mL (6) and 1 L)
test tubes (six)
stoppers to fit test tubes
graduated cylinder (10 mL)
dropper
ethanol [CH_3CH_2OH] (2 mL)
methanol [CH_3OH] (5 mL)
butan-1-ol [$CH_3CH_2CH_2CH_2OH$] (2 mL)
pentan-1-ol [$CH_3(CH_2)_4OH$] (2 mL)
octan-1-ol [$CH_3(CH_2)_7OH$] (2 mL)
3-methylbutan-1-ol (isoamyl alcohol) [$CH_3CH(CH_3)CH_2CH_2OH$] (2 mL)
salicylic acid (2 g)
acetic acid (glacial acetic acid) [CH_3COOH] (10 mL)
concentrated sulfuric acid [H_2SO_4] (2 mL)

Procedure

SAFETY NOTE:

- **Concentrated acetic and sulfuric acids are very corrosive and must be handled with extreme care.**
- **If any of the concentrated acetic or sulfuric acids comes in contact with your skin immediately wash the affected area with large quantities of water.**

1. Label six test tubes A, B, C, D, E and F and place the following reagents in each.

   Tube A: 2 mL of ethanol, 2 mL of acetic acid and 5 drops of concentrated H_2SO_4.
   Tube B: 2 mL of butan-1-ol, 2 mL of acetic acid, and 5 drops of concentrated H_2SO_4.

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Tube C: 2 mL of pentan-1-ol, 2 mL of acetic acid, and 5 drops of concentrated H_2SO_4.

Tube D: 2 mL of octan-1-ol, 2 mL of acetic acid, and 5 drops of concentrated H_2SO_4.

Tube E: 2 mL of 3-methylbutan-l-ol, 2 mL of acetic acid, and 5 drops of concentrated H_2SO_4.

Tube F: Salicylic acid crystals to a depth of about 1 cm, just enough methanol to dissolve the acid (about 2-3 mL), and 5 drops of concentrated H_2SO_4.

2. Stopper each test tube and set aside.

SAFETY NOTE:
- **Most organic substances are flammable and should be kept clear of naked flames.**
- **Turn off the Bunsen before heating the mixtures in a hot water bath to make the esters.**

3. Prepare a hot water bath by heating to boiling about 400 mL of tap water in a 1 L beaker.
4. When the water has boiled, turn off the Bunsen burner.
5. Remove the stoppers from the test tubes and place the test tubes in the water bath for 15 minutes.
6. Remove one of the test tubes from the water bath and pour the contents into a 100 mL beaker containing about 10 mL of tap water. Cautiously attempt to identify the odour of the ester floating on the water. Record your observations.
7. Pour the contents of each of the remaining test tubes, in turn, into a clean 100 mL beaker containing 10 mL of fresh tap water and note the odour. Use fresh water for each test.

Processing of results, and questions

1. Write equations for the formation of the esters you made
2. Write a general equation for the esterification reaction.
3. Write the names of each of the esters produced in this activity.
4. From the odours of the esters, identify where you have encountered any of them previously.
5. What is the role of the sulfuric acid in the esterification reactions?
6. What starting materials would you use to prepare ethyl butanoate, butyl ethanoate and methyl benzoate?
7. Why did the experiment call for glacial acetic acid and not for acetic acid solution?
8. Why did the experiment require you to pour the reactants and products into a beaker of water in step 7?